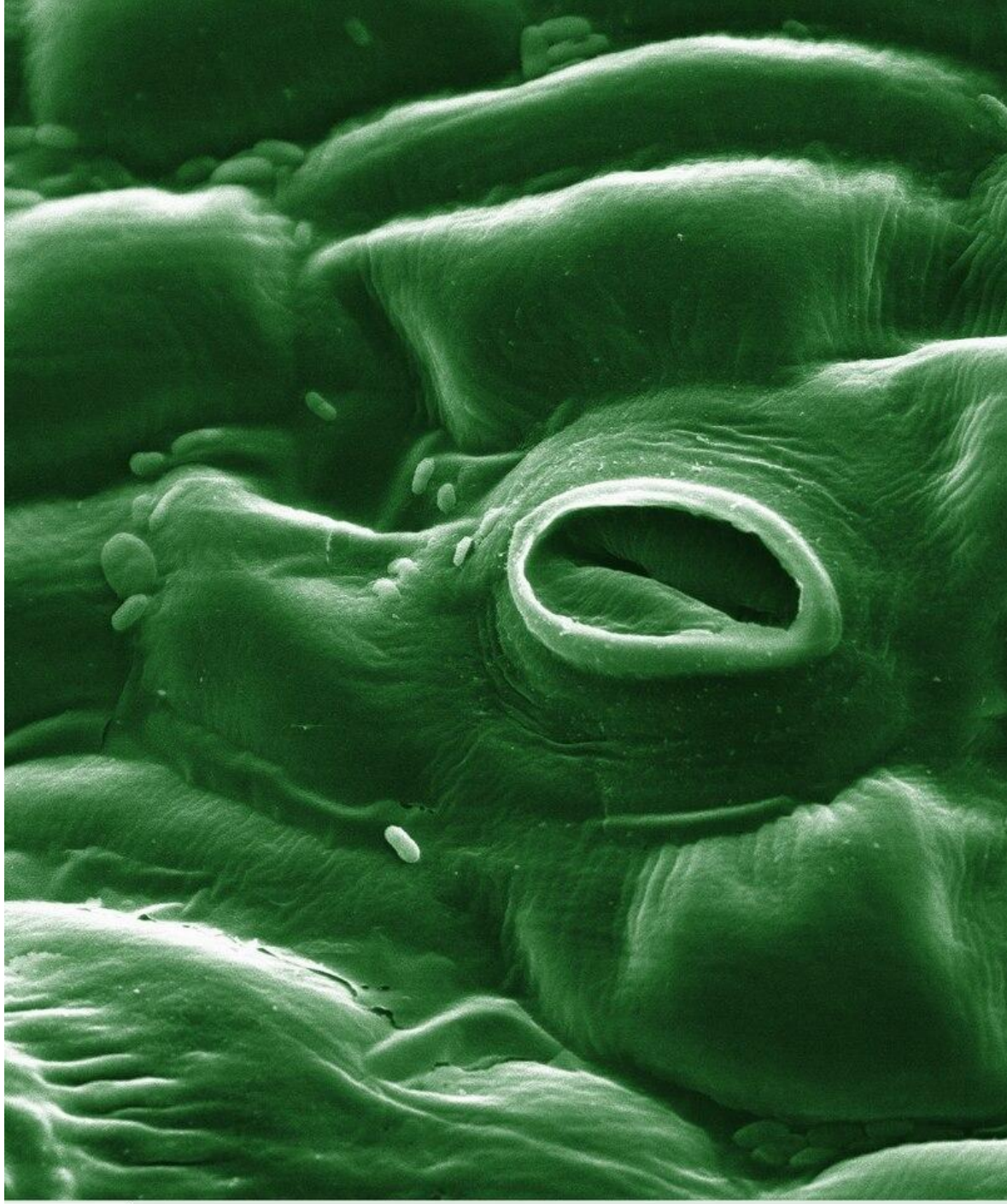
DP Biology – B3.1 Plant Systems

Gas exchange in plants

- What adaptations let multicellular organisms exchange gases?
- How do flowering plants and mammals differ, and resemble each other, in gas exchange?

Plant statements B3.1.7–10, SL and HL. Textbook pp. 283–291.
Image: a stoma on a tomato leaf, coloured electron micrograph. Photohound, public domain.

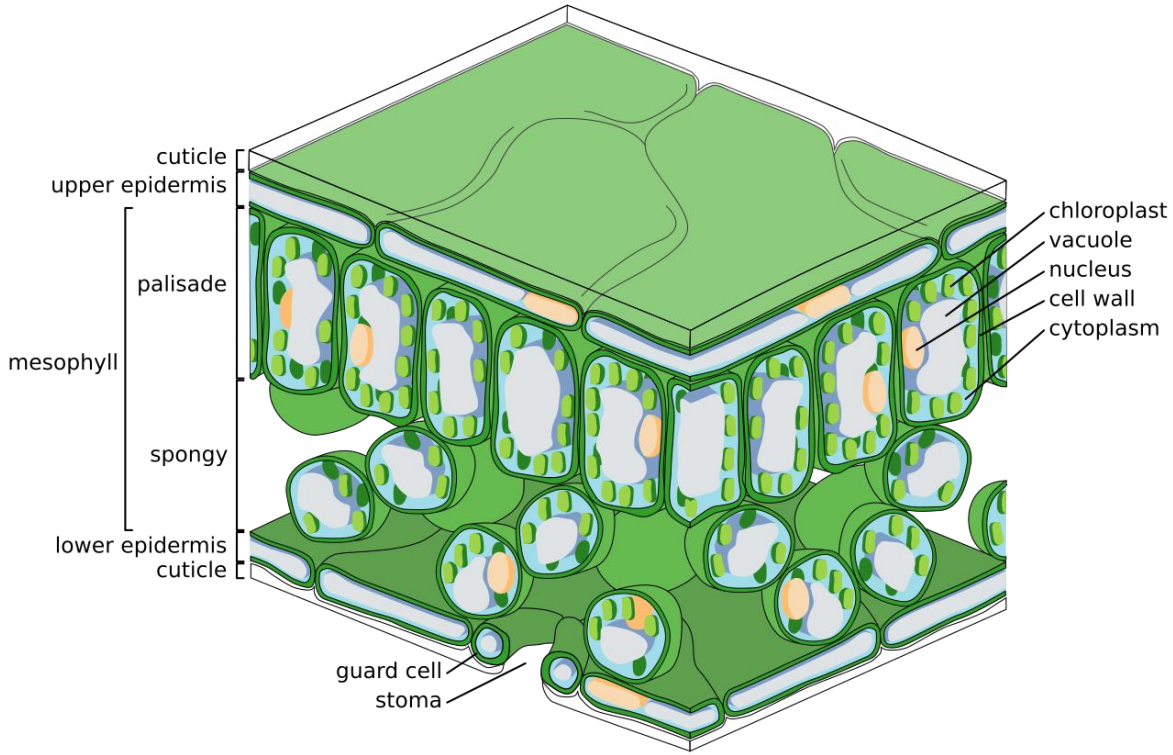


Objectives for each class

Class	Statements	By the end of the class, you can...
1	B3.1.7–8	Explain how leaf structures are adapted to gas exchange, and draw a plan diagram of a leaf
2	B3.1.9–10	Explain why gas exchange causes transpiration and what affects its rate, and determine stomatal density

Leaves need gas exchange too

B3.1.7



Tissues in a leaf. Zephyris, CC BY-SA 3.0.

Photosynthesis takes in **carbon dioxide** and releases **oxygen**. Gases diffuse through **air spaces** inside the leaf.

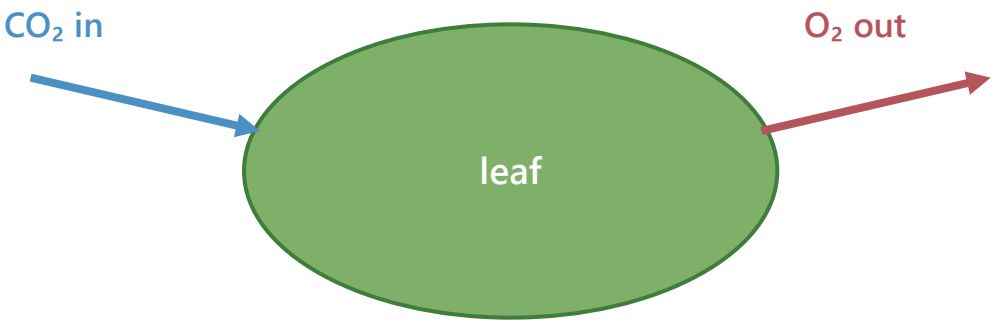
Leaves are **thin and flat**: a large surface and short diffusion distances.

Key idea

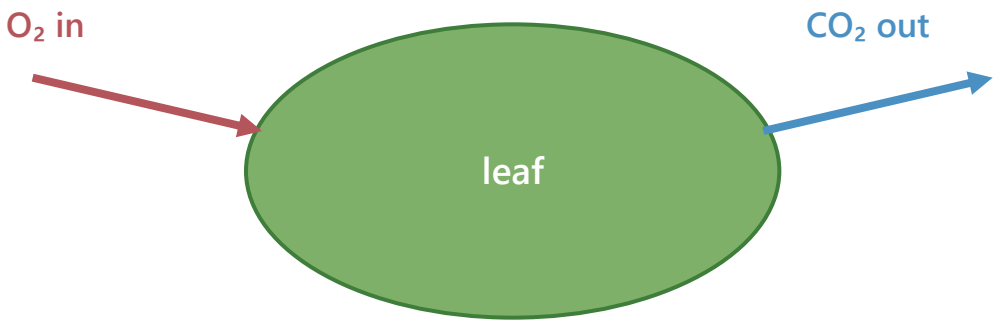
The pores that let carbon dioxide in also let water vapour out. Every leaf adaptation balances the two.

Gas exchange by day and by night

B3.1.7



Day: photosynthesis faster than respiration



Night: respiration only

Plant cells **respire all the time**. In the light, photosynthesis is usually faster, so there is a net uptake of CO_2.

Think

At dawn and dusk there is no net exchange of CO_2. Suggest why.

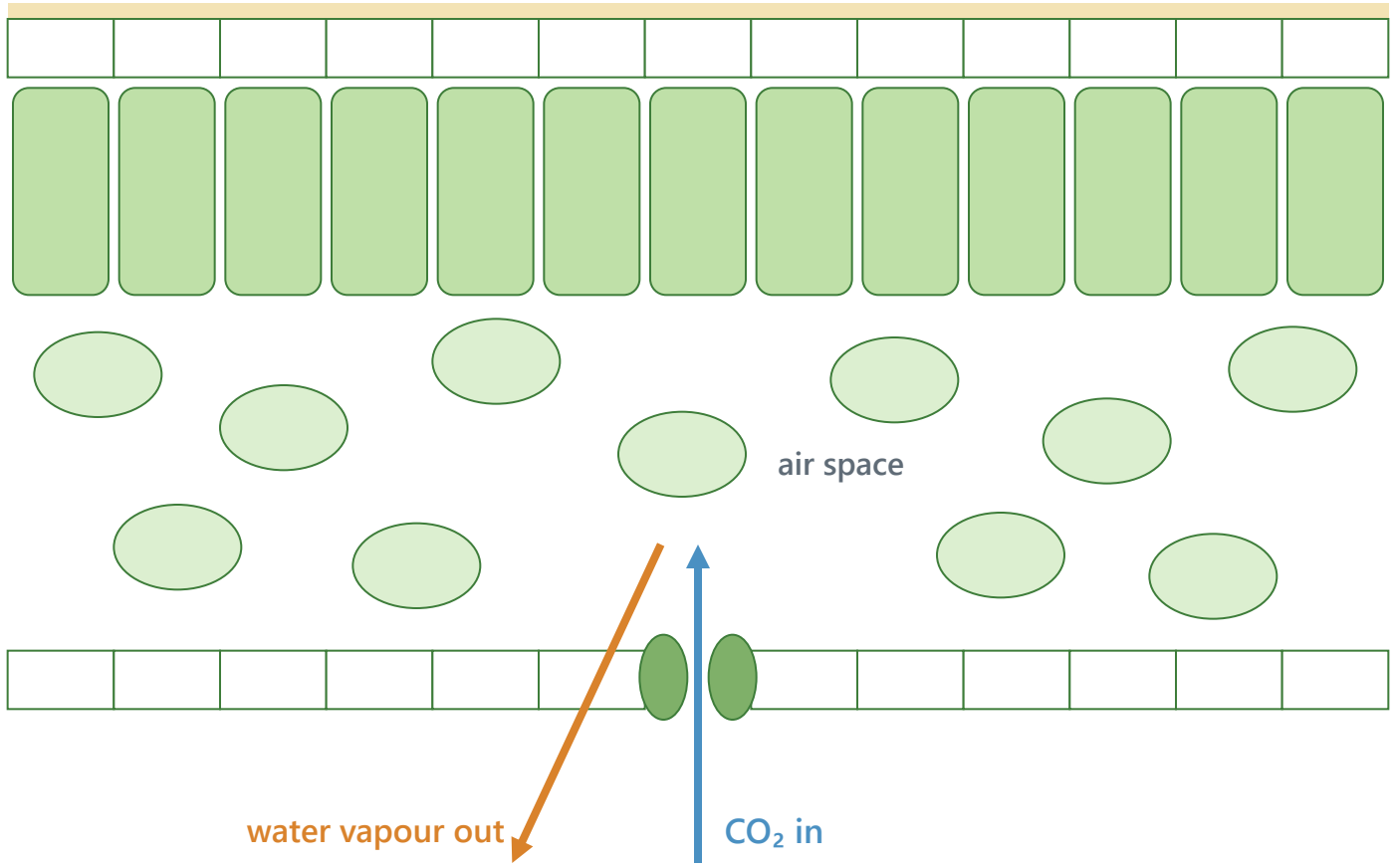
How leaf structures help

B3.1.7

Structure	Adaptation
Waxy cuticle	Waterproof: gases and water vapour must use the stomata
Epidermis	Thin, transparent; most stomata on the cooler, shaded lower side
Air spaces	Gases circulate around mesophyll cells
Spongy mesophyll	Large, moist surface for gases to dissolve and diffuse
Guard cells	Open and close stomata to control exchange and water loss
Veins	Xylem brings water; phloem removes sugar

Inside a leaf

B3.1.7



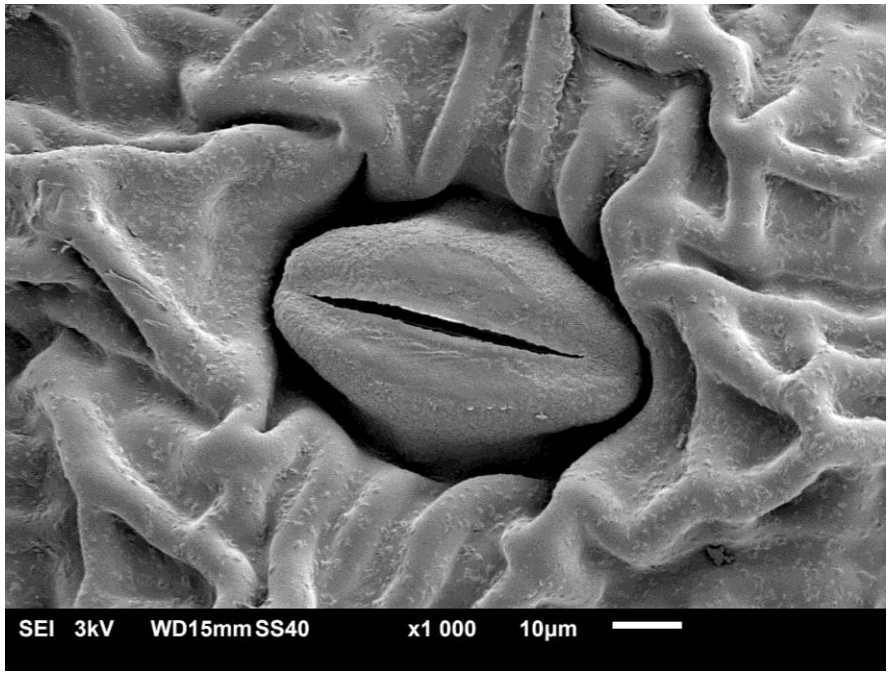
Drawn section, not to scale. Guard cells sit either side of the stoma in the lower epidermis.

CO_2 diffuses in through the **stoma**, through the **air spaces**, and dissolves in the moist walls of mesophyll cells.

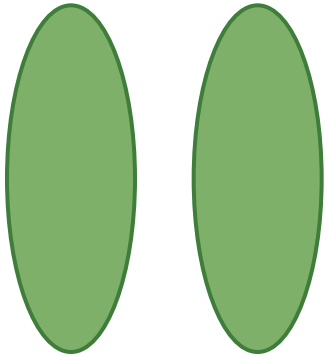
Palisade cells, packed with chloroplasts, do most photosynthesis.

Guard cells control the stoma

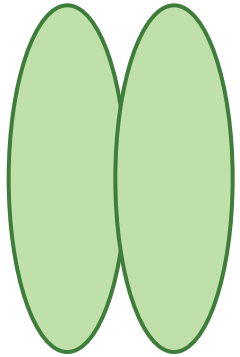
B3.1.7



Stoma and guard cells. Judyta Dulnik, CC BY-SA 4.0.



turgid: pore open



flaccid: pore closed

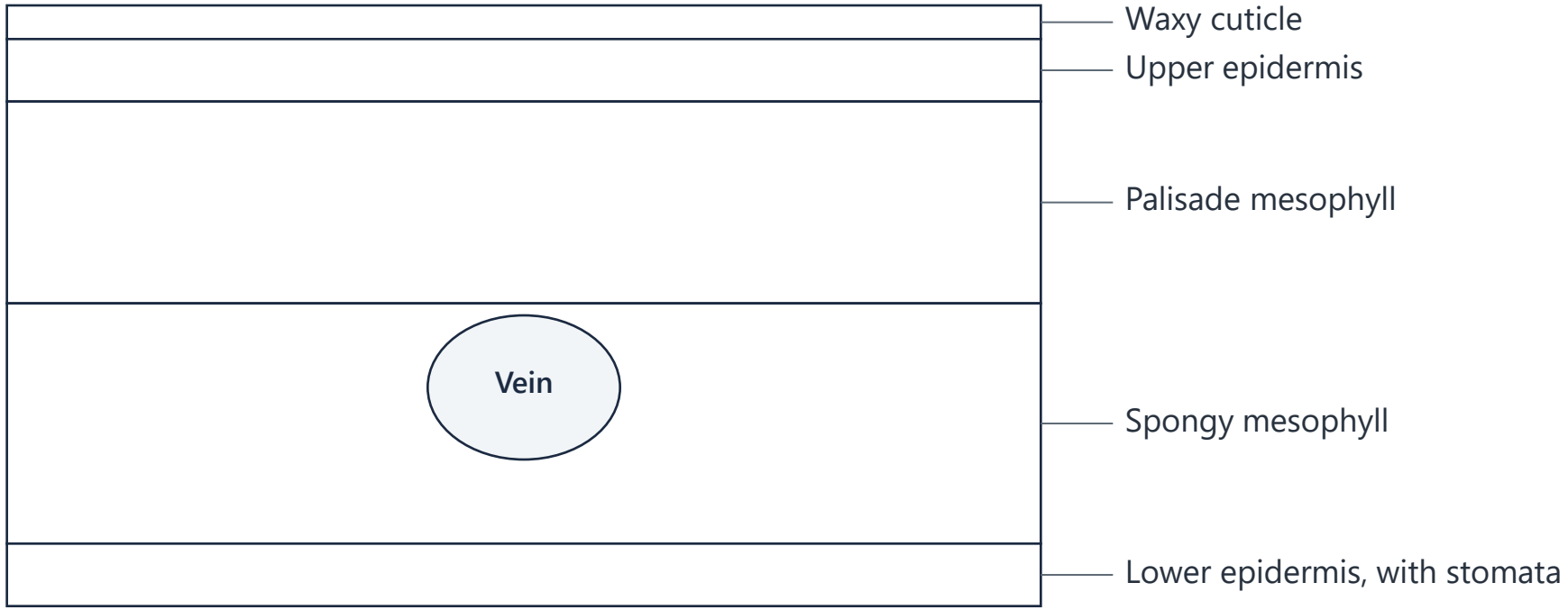
Water enters the guard cells, they become **turgid** and curve apart: the pore **opens**. When they lose water they become **flaccid** and it **closes**.

Common mistake

The stoma is the pore; guard cells are the two cells around it.

Plan diagram of a dicot leaf

B3.1.8



Schematic plan diagram of a transverse section.

Plan diagram

Show tissues as regions. No individual cells, no shading.

Activity

Draw a plan diagram from a micrograph of a leaf section. Use continuous lines, keep the proportions, and label cuticle, epidermis, palisade and spongy mesophyll, xylem and phloem.

Leaves and lungs compared

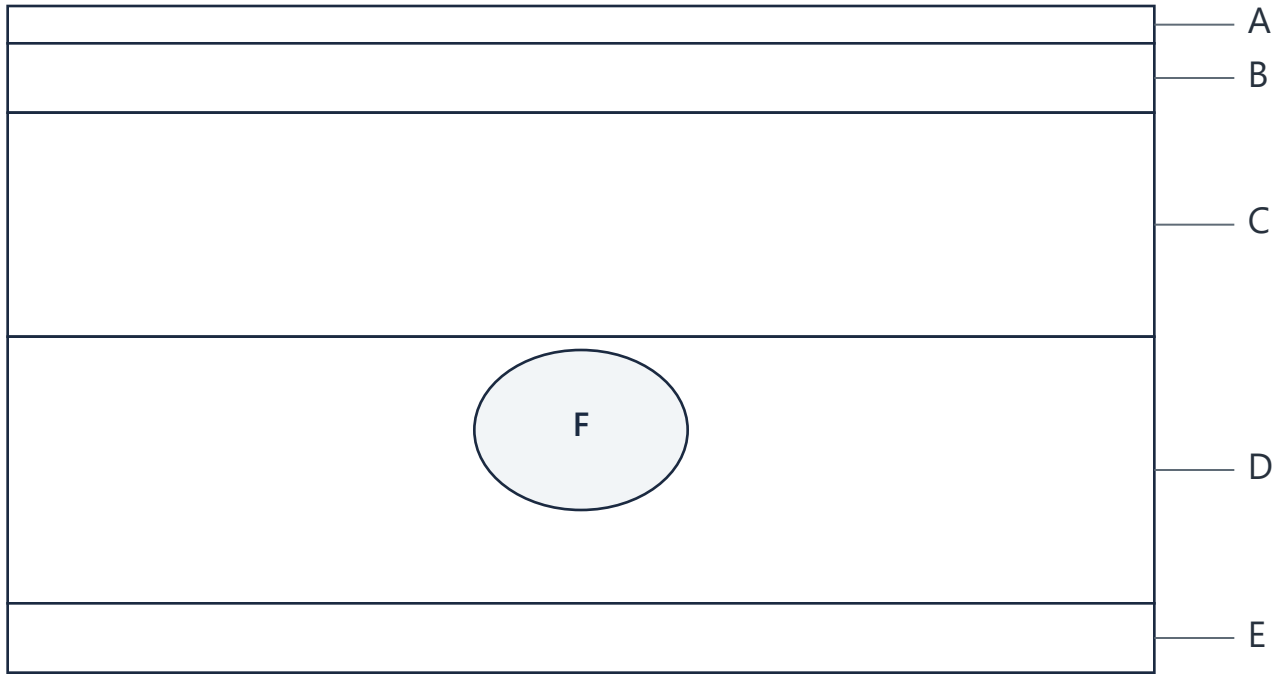
B3.1.7

	Mammal lung	Leaf
Exchange surface	Alveolar walls	Moist walls of mesophyll cells
Large area from	Many small alveoli	Air spaces around cells
Main gas taken in	Oxygen	Carbon dioxide (in light)
Gradient kept by	Ventilation, blood flow	Photosynthesis, air movement
Water loss limited by	Lungs deep inside the body	Waxy cuticle, guard cells

Link
This answers the second guiding question. The mammal side is B3.1.1–6.

Practice: name the tissues

B3.1.8



1. Name tissues A to F.
2. Which tissue has most chloroplasts?
3. In which tissue are most stomata found?
4. State one adaptation of D for gas exchange.

How to use this

Answer without looking back, then check against slide 8.

Check-in: class 1

B3.1.7–8

1. State the net gas exchange of a leaf in bright light and in the dark.
2. Give three features of a leaf that help gas exchange.
3. Explain how guard cells open a stoma.
4. State why a plan diagram has no individual cells.
5. Compare the exchange surface of a leaf and a lung.

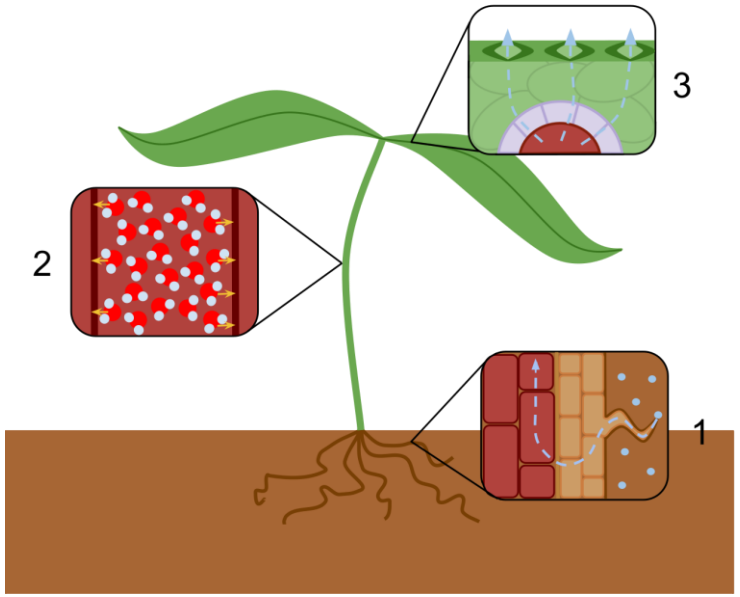
How to use this

Answer without your notes, then check against slides 3–10.

Transpiration follows from gas exchange

B3.1.9

Stomata open so CO_2 can enter → Water evaporates from moist cell walls → Water vapour fills the air spaces → **Vapour diffuses out through stomata**



Water moving through a plant. Laurel Jules, CC BY-SA 3.0.

This loss of water vapour from leaves is **transpiration**. It is the unavoidable cost of letting carbon dioxide in.

Common mistake

Water vapour diffuses out. It is not osmosis and not liquid water.

What affects the rate of transpiration

B3.1.9

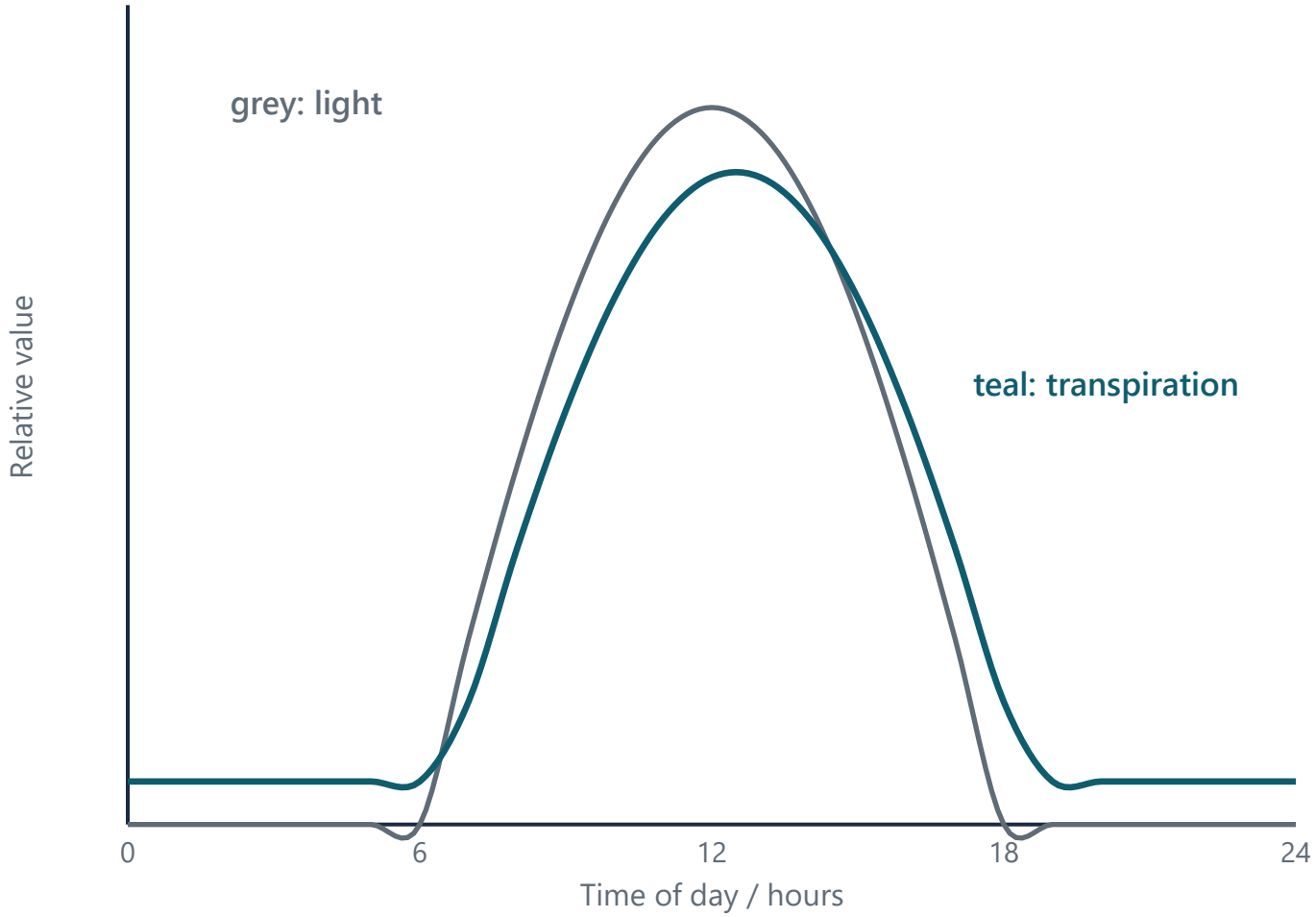
Factor	If it rises	Why
Temperature	Rate rises	More evaporation; steeper water vapour gradient
Humidity	Rate falls	Outside air holds more vapour: smaller gradient
Wind	Rate rises	Moist air near the stomata is swept away
Light	Rate rises	Stomata open for photosynthesis

Exam note

For every factor, explain using the concentration gradient of water vapour.

Working with data: transpiration over a day

B3.1.9

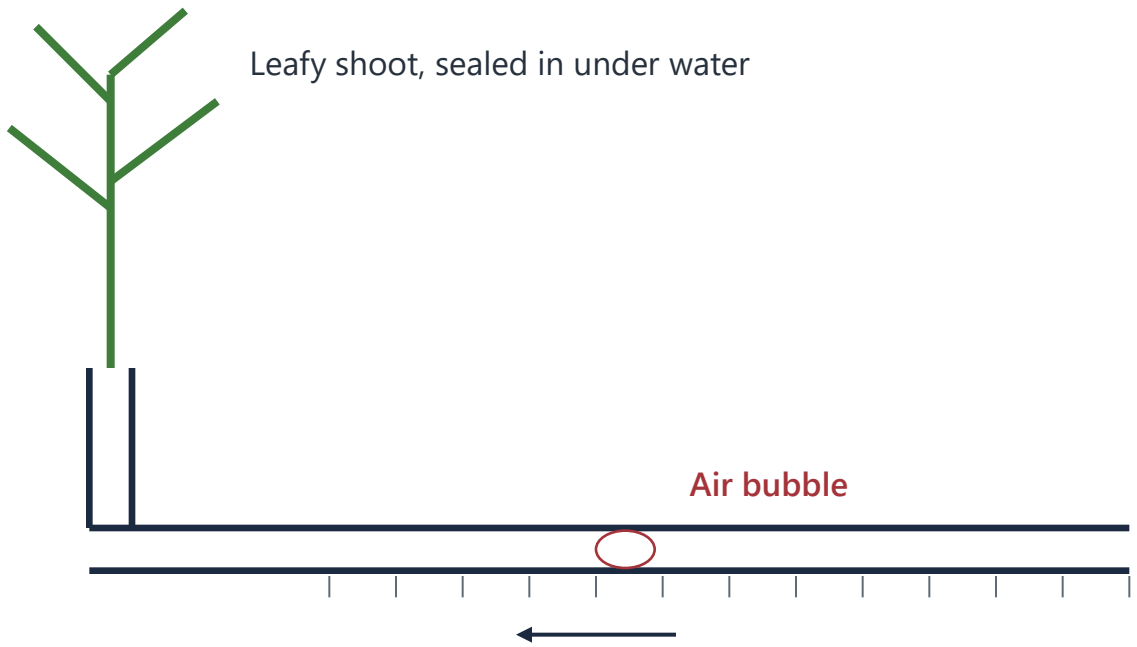


Invented values with a realistic pattern, for practice.

1. Describe how the rate changes over the day.
2. Explain why the rate peaks early in the afternoon.
3. Suggest why the rate is not zero at night.
4. Predict the curve on a humid day.

Measuring transpiration: the potometer

B3.1.9



Capillary tube and scale. The bubble moves towards the shoot.

Drawn, not to scale.

Water lost from the leaves is replaced by water drawn up the stem, so the bubble moves. **Rate = distance / time**.

Change one factor: a fan for wind, a plastic bag for humidity, a lamp for light.

Common mistake

A potometer measures water **uptake**, which is close to, but not exactly, water lost.

Working with data: a potometer experiment

B3.1.9

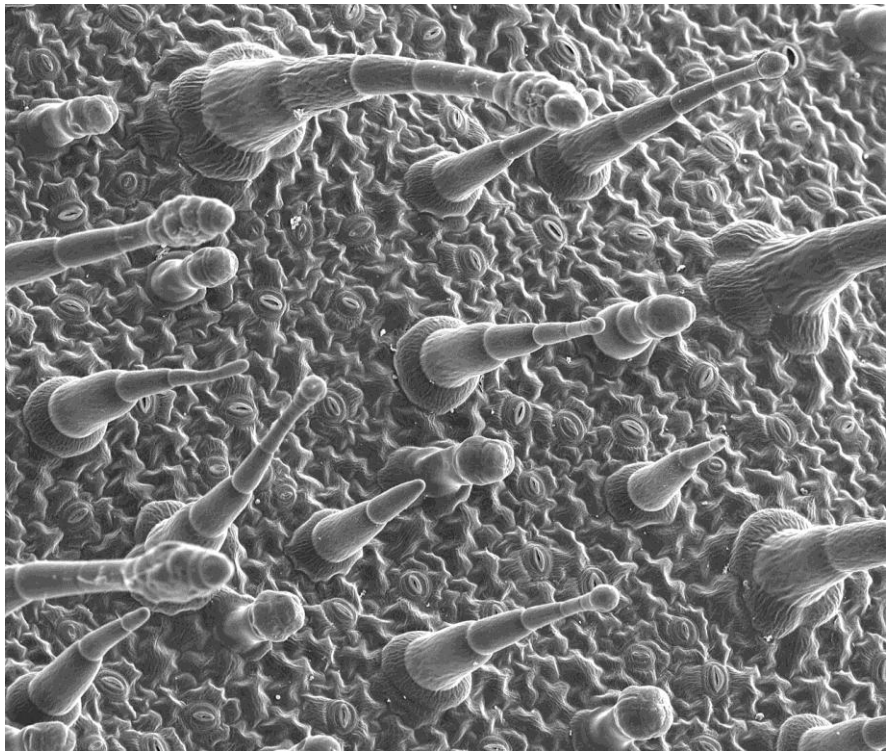
Distance moved by the bubble in 5 minutes. These data are invented for practice; the pattern matches real results.

Condition	Distance / mm	Rate / mm min^{-1}
Still air, room light	20	?
Fan on	45	?
Plastic bag over shoot	6	?
Lamp on	34	?

1. Calculate each rate.
2. Explain the result with the fan.
3. Explain the result with the bag.
4. State one assumption of the method.

Stomatal density

B3.1.10



Leaf epidermis. Louisa Howard, Dartmouth, public domain.

Stomatal density is the number of stomata per unit area of leaf.

Use a micrograph, or a **leaf cast**: paint clear nail varnish on the leaf, let it dry, lift it with clear tape, and count stomata in several fields of view.

Formula

Area of field of view = 3.14 × r^2. Density = mean count / area.

Worked example: calculating density

B3.1.10

Field of view	Stomata counted
1	22
2	18
3	20

Invented values, for practice.

1. Mean count = (22 + 18 + 20) / 3 = 20
2. Field diameter 0.4 mm, so r = 0.2 mm
3. Area = 3.14 × 0.2^2 = 0.126 mm^2
4. Density = 20 / 0.126 = 159 per mm^2

Activity

Make casts of the upper and lower surfaces of one leaf. Calculate both densities and explain the difference.

Practice: comparing leaves

B3.1.10

Stomatal density of two species. The values are invented for practice; the pattern is realistic.

Species and habitat	Upper / mm^{-2}	Lower / mm^{-2}
X: shady, damp woodland	20	160
Y: dry, sunny hillside	0	90

1. Compare the two surfaces of species X.
2. Compare the two species.
3. Explain how Y's density suits its habitat.
4. Calculate the ratio lower : upper for X.

Check-in: class 2

B3.1.9–10

1. Explain why transpiration is a consequence of gas exchange.
2. State the effect of humidity on transpiration, and explain it.
3. Outline how a potometer is used.
4. Define stomatal density.
5. A mean of 15 stomata are counted in 0.15 mm^2. Calculate the density.

How to use this

Answer without your notes, then check against slides 12–19.

Common mistakes to avoid

Not this	But this
Plants do not respire.	Plants respire all the time; by day photosynthesis is faster.
Transpiration is evaporation from the leaf surface.	Water evaporates inside the leaf and diffuses out through stomata.
Guard cells open by losing water.	They open when they gain water and become turgid.
A potometer measures transpiration exactly.	It measures water uptake.

Summary: gas exchange in plants

B3.1.7–10

	Key idea
B3.1.7	Thin leaf, air spaces, moist mesophyll, stomata; cuticle limits water loss
B3.1.8	Plan: cuticle, epidermis, palisade, spongy mesophyll, vein, epidermis
B3.1.9	Open stomata let water vapour out; rate set by the vapour gradient
B3.1.10	Density = stomata per mm^2, from casts or micrographs

Check that you can...

1. **Explain** how leaf structure suits gas exchange
2. **Draw** and **label** a plan diagram of a dicot leaf
3. **Explain** how stomata open and close
4. **Explain** how temperature, humidity, wind and light affect transpiration
5. **Calculate** stomatal density and potometer rates
6. **Compare** gas exchange in leaves and lungs

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain why gas exchange in a leaf leads to water loss.
2. Predict the effect of higher humidity on the rate of transpiration, and explain why.
3. Name two tissues in a plan diagram of a leaf and give one adaptation of each.

Image credits

Tomato leaf stomate 1-color.jpg – Photohound – Public domain
Leaf Tissue Structure.svg – Zephyris – CC BY-SA 3.0
Stoma.jpg – Judyta Dulnik – CC BY-SA 4.0
Transpiration Overview.svg – Laurel Jules – CC BY-SA 3.0
Leaf epidermis.jpg – Louisa Howard (Dartmouth electron microscope facility) – Public domain

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